

RETAILSPHERE (PTY) LTD

BPA-001 — Stakeholder Interview Reference

Interview 2 — Zanele Mokoena

Full Name	Zanele Mokoena
Title	National Inventory Manager
Domain	Inventory Management
Interview Date	17 April 2026
Interview Status	Complete
ADRs Triggered	ADR-001 (Master Product ID)
BPA Finding	BPA-FINDING-001 — Structural Data Integrity Risk

1. Process Owner Profile

Zanele Mokoena is the National Inventory Manager at RetailSphere, responsible for stock management across 87 stores and one distribution centre. She is precise, data-driven, and has been managing inventory without reliable real-time visibility for years. She brought her manually maintained Monday morning replenishment spreadsheet into the interview — three hours of work every week that the platform must replace. Her problems are structural, not process-driven, which makes them harder to solve and more important to acknowledge formally before the platform goes live.

2. Key Business Processes

The disease, not the symptom — Zanele's core problem:

Zanele cannot tell you reliably how many units of any SKU exist in any store right now. Three data gaps undermine every inventory metric:

- Shrinkage — captured weekly on paper or Excel per store, emailed to Zanele's team every Monday. Not integrated into any replenishment system. Not real-time.
- Inter-store transfers — captured in SAP eventually. Sometimes two days after the transfer occurred. System stock position is outdated.
- Returns — POS captures customer returns as negative sales. Damaged returns that go back to the DC are captured separately — or not at all.

The replenishment process:

- Store manager identifies low stock — based on an unreliable system position.
- Replenishment order raised to DC — quantity based on unreliable data.
- DC picks stock and dispatches to store.

- Store receives and counts stock.

Zanele currently manages 340 open replenishments manually in a spreadsheet rebuilt every Monday morning — three hours of work.

3. Decisions Made — Daily, Weekly, Monthly

Daily

- Monitor open replenishment orders — currently via spreadsheet.
- Respond to store manager queries about stock availability.
- Manage DC picking priorities.

Weekly

- Rebuild replenishment tracking spreadsheet — 3 hours every Monday.
- Review shrinkage reports emailed from 87 stores.
- Identify which stores have highest variance between system stock and physical count.

Monthly

- Inventory reporting to Nomvula — stock levels, shrinkage rates, replenishment cycle times.
- Identify product categories with highest stock-out risk.

4. Questions Zanele Cannot Currently Answer

- How many units of SKU X do we have in store Y right now — with confidence?
- Which stores have the biggest gap between system stock and physical stock?
- Which product categories have the highest shrinkage rates?
- How long does a replenishment order take from request to store receipt — by store, by DC, by category?
- Where in the replenishment pipeline are the 340 open orders right now?

5. What Good Looks Like in 12 Months

Zanele's manual Monday morning spreadsheet replaced by automated dashboard:

- Open replenishment orders by store, by SKU, by pipeline stage — real-time.
- Average days in replenishment pipeline by store, DC, and product category.
- Stores with biggest variance between system stock and physical count.
- Product categories with highest shrinkage rates.
- Inter-store transfer cycle times.

Zanele's exact words:

"When a store manager in Polokwane asks me how many units of a jacket we have in his store right now — I cannot answer that question with confidence."

— Zanele Mokoena, National Inventory Manager

6. Data Sources and Quality Issues

Source Systems

- SAP ECC 6.0 — central stock management, warehouse movements, replenishment triggers.
- Oracle MICROS POS — real-time stock decrements at point of sale across 87 stores. Schema varies per store — 87 different database schemas.

Fact Table Design — Two Tables Required

Inventory Periodic Snapshot:

- One row per SKU per store per day.
- Grain: SKU + Store + Date.
- Key measure: Quantity on Hand — semi-additive. Cannot be summed across dates. Only averaged.

Replenishment Accumulating Snapshot:

- One row per replenishment order — updated at each pipeline stage.
- Grain: One row per replenishment order.
- Pipeline stages: Reorder Date → DC Pick Date → Dispatch Date → Store Receipt Date.

Accumulating Snapshot Column Classification

From Zanele's spreadsheet — classified for the dimensional model:

- Replenishment ID — Degenerate Dimension. Transaction identifier on fact table.
- Store — Foreign Key → Store Dimension. Conformed across Sales and Inventory.
- SKU — Foreign Key → Product Dimension. Resolves via ADR-001.
- Reorder Date — Foreign Key → Date Dimension (role-played as Reorder Date Key).
- DC Pick Date — Foreign Key → Date Dimension (role-played). Unknown on row creation → TBD date row.
- Dispatch Date — Foreign Key → Date Dimension (role-played). Unknown until DC dispatches.
- Store Receipt Date — Foreign Key → Date Dimension (role-played). Unknown until store confirms.
- Days in Pipeline — Stored Measure. ETL calculates. Includes weekend exclusion and timestamp validation. Stored not derived per Kimball Chapter 6.
- Status — Derived in view. Not stored. Calculated from date columns. Open / Picked / In Transit / Received.

Unknown Date Problem — Important Design Note

When a replenishment row is first created only Reorder Date is known. Three date keys are unknown. Per Kimball Chapter 6 the Date dimension needs a special row for 'To Be Determined' — surrogate key -1 by convention. New replenishment rows start with three date keys pointing to this row.

BPA-FINDING-001 — Inventory Domain Structural Data Integrity Risk

The Inventory Management domain data quality problems are structural — not process-driven. Replenishment trigger, shrinkage capture, and returns tracking are broken at the operational source level.

The Gold layer will produce inventory metrics that are directionally correct but quantitatively approximate until three upstream operational gaps are resolved:

1. Real-time stock position visibility at store level.
2. Integration of shrinkage data into a governed system.
3. Unified returns capture across POS and DC.

This finding must be formally acknowledged by the business before the platform goes live. Contrast with Procurement: Procurement problems are process-driven and fixable. Inventory problems are structural and require operational change beyond the data platform scope.

7. Architectural Implications

- Two fact tables required: Inventory Periodic Snapshot and Replenishment Accumulating Snapshot.
- Date dimension needs a special TBD row (surrogate key -1) for unknown pipeline stage dates.
- ADR-001 dependency: SKU in replenishment must resolve to same Product dimension surrogate key as Sales and Procurement domains.
- Conformed Store dimension: Same Store dimension used in Sales, Inventory Periodic Snapshot, and Replenishment Accumulating Snapshot.
- Status derived in view not stored: Reduces inconsistency risk given 87-store data quality variation.
- Cross-domain GRN dependency: GRN timestamp fix (Bronze layer four columns) directly improves Inventory data quality even though GRN is captured in Procurement domain.

8. Key Quote

"I manage inventory across 87 stores and one distribution centre. Every night each store syncs its sales transactions to central. That tells me what was sold. But it does not tell me reliably what is on the shelf right now."

— Zanele Mokoena, National Inventory Manager

— End of Interview Reference — Zanele Mokoena